

2017EBN122C3**VRAESTEL**

Kopiereg voorbehou

October 2017

2017EBN122C3**QUESTION PAPER**

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October 2017

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		8 + OCR	

Important

- The examination regulations of the University of Pretoria apply.
- All questions must be answered on the Optical Character Recognition (OCR) form.
- Read the instructions on the OCR form carefully.
- Answers must include units, and must be written as rational numbers.

Dosent(e) / <i>Lecturer(s):</i>	D le Roux, D Ramotsoela, X Ye
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Do step 0 before the start of the session.**Do step 1 in the first 30 minutes of the test****Do step 2 within the following 10 minutes.***Memo.***STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN122C3**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

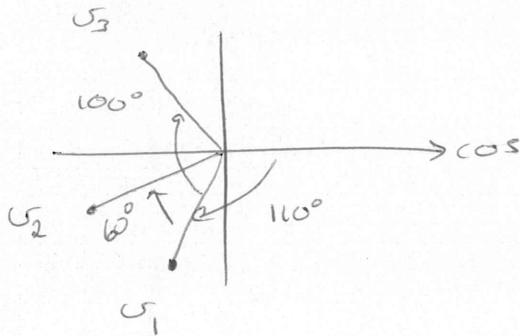
- Complex numbers:Rectangular form: always place the "j" **at the start** of the imaginary part of the complex number.

Polar form: always add the "°" (degree sign) after the angle, and make a clear angle sign.

Examples

01	-	1	.	4	+	j	8			μ	A						
02	3	4	0	∠	4	5	9			V							
03	-	j	5	4						K	Ω						
04	1	∠	2	0	°												
05	5	0	∠	-	1	3	0	°		m	A						

Questions 01-02 [2]		Vraag 01-02 [2]	
The following sinusoid is given: $v_1 = 16 \cos(\omega t - 110^\circ) \text{ V}$ It is known that - v_2 lags v_1 by 60°. - v_1 leads v_3 by 100°.		Die volgende sinusied word gegee: $v_1 = 16 \cos(\omega t - 110^\circ) \text{ V}$ Dit is bekend dat: - v_2 loop na v_1 met 60°. - v_1 loop voor v_3 met 100°.	
01	What is the phase angle ϕ_d between v_2 and v_3 ? ($0 \leq \phi_d \leq 180^\circ$)	01	Wat is die fasehoek ϕ_d tussen v_2 en v_3 ? ($0 \leq \phi_d \leq 180^\circ$)
02	Choose the correct statement: 1 v_2 leads v_3 2 v_3 leads v_2 (Write 1 for statement 1, and 2 vir statement 2.)	02	Kies die korrekte stelling: 1 v_2 loop voor v_3 2 v_3 loop voor v_2 (Skryf 1 vir stelling 1, en 2 vir stelling 2.)

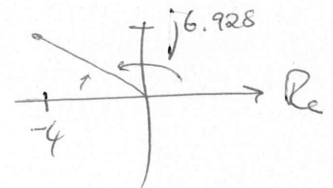


$$\phi_d = 100^\circ - 60^\circ = 40^\circ \checkmark$$

Statement 1 is correct.

Questions 03 [2]		Vraag 03 [2]	
03	Solve for z : $z = \frac{14.142 \angle 40^\circ}{\sqrt{-4 + j6.928}}$ (Write the final answer in polar form .)	03	Los op vir z : $z = \frac{14.142 \angle 40^\circ}{\sqrt{-4 + j6.928}}$ (Skryf die finale antwoord in polêre vorm .)

$$z = \frac{14.142 \angle 40^\circ}{\sqrt{4^2 + 6.928^2} \angle 180^\circ - \tan^{-1}\left(\frac{6.928}{4}\right)}$$



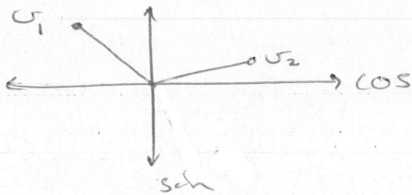
$$= \frac{14.142 \angle 40^\circ}{(8 \angle 120^\circ)^{1/2}}$$

$$= \frac{14.142 \angle 40^\circ}{\sqrt{8} \angle 120^\circ / 2}$$

$$= \frac{14.142 \angle 40^\circ}{2\sqrt{2} \angle 60^\circ}$$

$$= 5 \angle -20^\circ$$

Question 04 [2]	Vraag 04 [2]
The following sinusoids are given: $v_1 = -5.1 \cos(\omega t - 53.97^\circ) \text{ V}$ $v_2 = 6.088 \sin(\omega t + 100.2^\circ) \text{ V}$ Determine the phasor value of $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Give the answer in polar form.	Die volgende sinusiede word gegee: $v_1 = -5.1 \cos(\omega t - 53.97^\circ) \text{ V}$ $v_2 = 6.088 \sin(\omega t + 100.2^\circ) \text{ V}$ Bepaal die fasor waarde van $\bar{V}_T = \bar{V}_1 + \bar{V}_2$. Gee die antwoord in polêre vorm.
04 $\bar{V}_T$	04 $\bar{V}_T$



$$v_1 = -5.1 \cos(\omega t - 53.97^\circ)$$

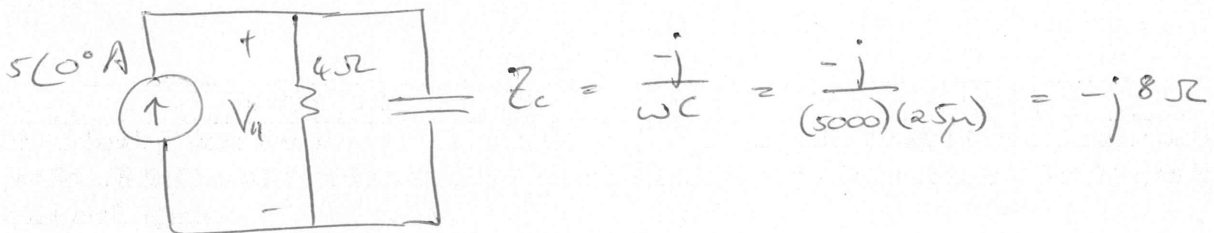
$$= 5.1 \cos(\omega t + 126.03^\circ) \rightarrow \bar{V}_1 = 5.1 \angle 126.03^\circ$$

$$v_2 = 6.088 \sin(\omega t + 100.2^\circ)$$

$$= 6.088 \cos(\omega t + 10.2^\circ) \rightarrow \bar{V}_2 = 6.088 \angle 10.2^\circ$$

$$\bar{V}_T = \bar{V}_1 + \bar{V}_2 = 6 \angle 60.1^\circ \checkmark$$

Question 05 [2]	Vraag 05 [2]
The following three elements are all connected in parallel: a $5 \cos(5000t) \text{ A}$ current source, a 4Ω resistor, and a $25 \mu\text{F}$ capacitor. What is the total voltage across the parallel combination of elements? (Give the answer in rectangular format.)	Die volgende drie elemente word in parallel gekoppel: 'n $5 \cos(5000t) \text{ A}$ stroombron, 'n 4Ω resistor, en 'n $25 \mu\text{F}$ kapasitor. Wat is die totale spanning oor die parallel kombinasie van elemente? (Give the answer in rectangular format.)
05 $\bar{V}_{ }$	05 $\bar{V}_{ }$

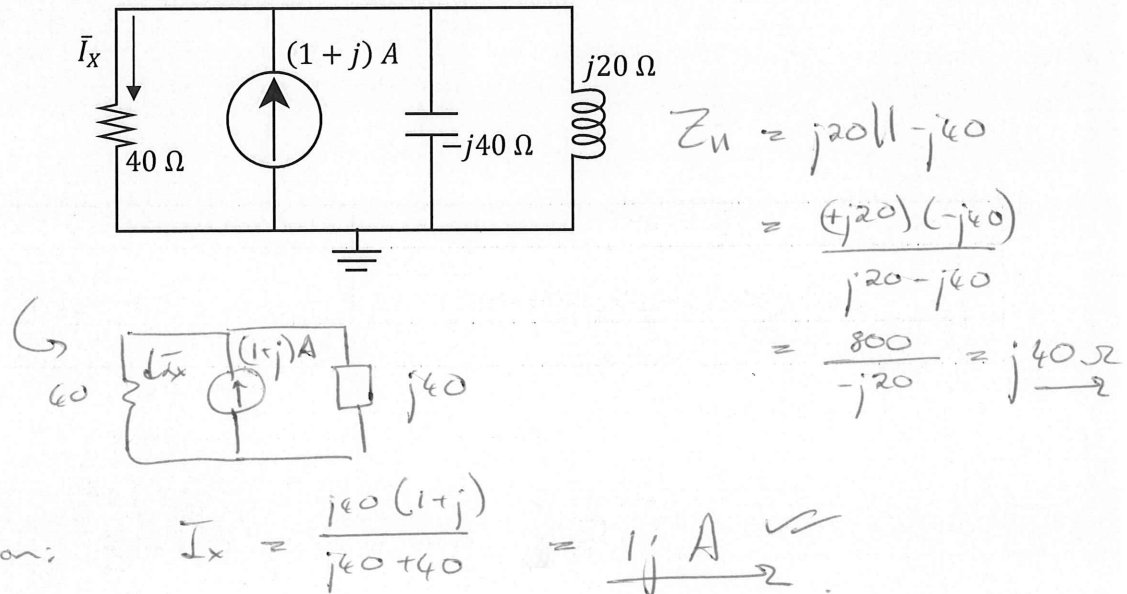


$$Z_n = 4 \parallel (-j8) = \frac{4 \times (-j8)}{4 - j8} = 3.2 - 1.6j \Omega$$

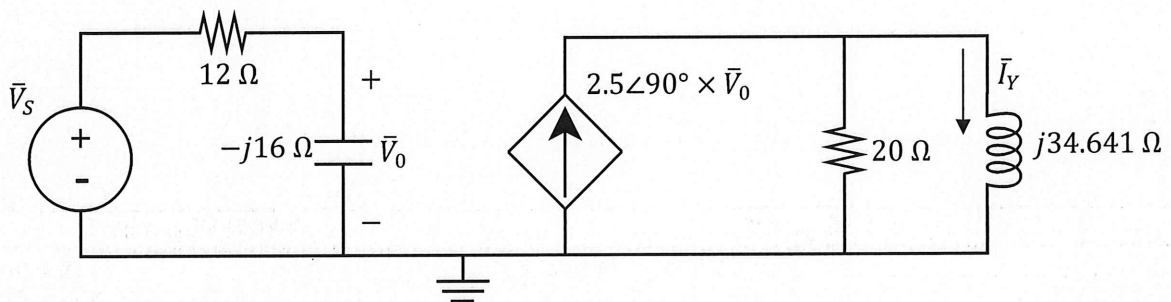
$$V_{||} = 5 \angle 0^\circ \times Z_n = 5 (3.2 - 1.6j)$$

$$= 16 - 8j \text{ V} \checkmark$$

Questions 06 [2]		Vraag 06 [2]	
Calculate $\bar{I}_X$. Give the final answer in rectangular form .		Bereken $\bar{I}_X$. Gee die finale antwoord in reghoekige vorm .	
06	$\bar{I}_X$	06	$\bar{I}_X$



Question 07 & 08 [4]		Vraag 07 & 08 [4]	
Given: $\bar{V}_0 = 4 \text{ V}$. Determine $\bar{V}_S$ and $\bar{I}_Y$.		Gegee: $\bar{V}_0 = 4 \text{ V}$. Bepaal $\bar{V}_S$ en $\bar{I}_Y$.	
07	$\bar{V}_S$ (Rectangular form)	07	$\bar{V}_S$ (Reghoekige vorm)
08	$\bar{I}_Y$ (Polar form)	08	$\bar{I}_Y$ (Polêre vorm)



* V-division: $V_0 = \frac{-j16 \times V_S}{12-j16}$

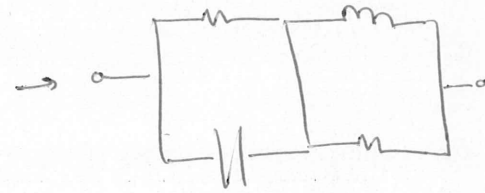
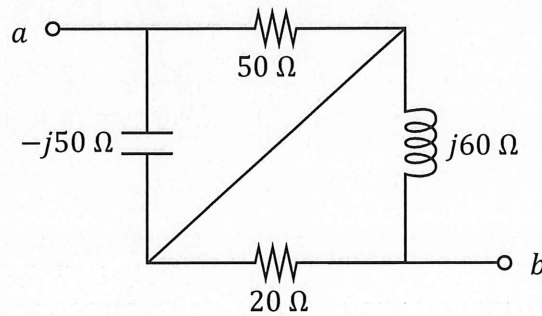
$\therefore V_S = \frac{4 \times (12-j16)}{-j16} = 4+j3 \text{ V} \checkmark$

* I-division: $I_Y = \frac{I_S \times 20}{20+j34.641}$

$$= \frac{(2.5 \angle 90^\circ \times 4) \times 20}{20+j34.641}$$

$$= 5 \angle 30^\circ \text{ A} \checkmark$$

Question 09 [2]		Vraag 09 [2]	
09	Calculate Z_{ab} . Give the answer in rectangular form .	09	Bepaal Z_{ab} . Gee die antwoord in reghoekige vorm .



$$Z_{ab} = (50 \parallel -j50) + (20 \parallel j60)$$

$$= \frac{50 \times (-j50)}{50 - j50} + \frac{20 \times j60}{20 + j60}$$

$$= 25 - j25 + 18 + j6$$

$$= 43 - j19\ \Omega$$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2

0

1

7

E

B

N

1

1

1

C

3

Remember:

Write a complex number in:

- rectangular form as $a \pm j b$ where j is at the **start** of the imaginary number.
- polar form as $a \angle \theta^\circ$ where the angle sign $\angle$ is clear, and you add the degree unit after the angle).

(See examples on page 1.)

	Answer	Unit
01	$\phi_d = 40$	$^\circ$
02	Statement #: 1	No unit required
03	$z = 5 \angle -20^\circ$ (polar)	
04	$\bar{V}_T = 6.001 \angle 60.097^\circ$ (polar)	V
05	$\bar{V}_{II} = 16 - j 8$ (rectangular)	V
06	$\bar{I}_X = j 1$ (rectangular)	A
07	$\bar{V}_S = 4 + j 3$ (rectangular)	V
08	$\bar{I}_Y = 5 \angle 30^\circ$ (polar)	A
09	$Z_{ab} = 43 - j 19$ (rectangular)	Ω